Blessed G/Michael Catholic Secondary School

Semester 1 Physics Final Examination for Grade10

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ **Grade & Sec:** Grade10 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** Specific gravity has no physical units.

____ **2.** Lightning is a natural example of massive electrical discharge.

____ **3.** A body becomes positively charged when it loses electrons.

____ **4.** The relative velocity of two bodies moving with the exact same velocity is zero.

____ **5.** A body moving with constant velocity has the same instantaneous and average velocity.

____ **6.** The area under an acceleration versus time graph is equal to displacement.

____ **7.** When mass of a given substance is doubled, its density also doubles.

____ **8.** The Young's modulus of a substance will be doubled when the stress exerted on it is doubled.

____ **9.** When a body is charged by conduction, it always gains electrons.

____ **10.** A neutral conductor can be charged by induction without touching a charged object.

II. Choose the best answer among the given alternatives.

____ **11.** Which one of the following statements correctly describes elastic materials?

A. Elastic materials regain their original shape after deformation when the deforming force is removed
B. Plastic materials can recover their original size or shape after being distorted
C. Plastic materials change their shape only by pressure
D. Elastic materials cannot recover their original size or shape after being distorted

____ **12.** A wire of cross-sectional area 2 * 10^-6 m^2 is stretched by a force of 50 N. The tensile stress developed in the wire is:

A. 1 * 10^7 Pa **B.** 2.5 * 10^7 Pa **C.** 5 * 10^7 Pa **D.** 1 * 10^6 Pa

____ **13.** The SI unit of electric charge is the:

A. Ampere (A) **B.** Tesla (T) **C.** Volt (V) **D.** Coulomb (C)

____ **14.** A wire of original length 1 m stretches by 1 mm under an applied tensile force. The tensile strain produced is:

A. 0.001 **B.** 0.01 **C.** 0.1 **D.** 0.0001

____ **15.** What is the mass of a 10 L volume of pure water? (Density of water = 1000 kg/m^3 = 1 kg/L)

A. 10 kg **B.** 0.1 kg **C.** 100 kg **D.** 0.01 kg

____ **16.** If the cross-sectional area of a stretched wire is doubled while keeping the stretching force constant, the developed stress is:

A. Doubled **B.** Halved **C.** Quartered **D.** One-fourth

____ **17.** A wire 2 m long with cross-sectional area 1 * 10^-6 m^2 stretches by 0.1 mm under a 100 N force. Its Young's modulus is:

A. 2 * 10^11 Pa **B.** 2 * 10^10 Pa **C.** 2 * 10^12 Pa **D.** 2 * 10^9 Pa

____ **18.** For the same applied stress, a material with a higher Young's modulus will experience:

A. More strain **B.** Less strain
C. The same strain **D.** Cannot be determined

____ **19.** An object of mass 3 kg hangs from three light inextensible wires in equilibrium. Two wires make angles theta1 = 53° and theta2 = 37° with the ceiling. What are tensions T1 and T2? (g = 10 m/s^2)

A. T1 = 18 N and T2 = 24 N **B.** T1 = T2 = 18 N
C. T1 = T2 = 24 N **D.** T1 = 24 N and T2 = 18 N

____ **20.** Uniformly accelerated rectilinear motion means that:

A. Acceleration is constant
B. Velocity is constant
C. Displacement is constant
D. Acceleration changes linearly with time

____ **21.** What is the density of a 2.7 kg uniform solid cube of side length 10 cm?

A. 2.7 kg/m^3 **B.** 27 kg/m^3 **C.** 270 kg/m^3 **D.** 2700 kg/m^3

____ **22.** Which one of the following physical laws best describes the behavior of elastic materials within their proportionality limit?

A. Gay-Lussac's law **B.** Hooke's law

C. Newton's second law **D.** Newton's law of universal gravitation

____ **23.** A steel wire (Y = 2 * 10^11 Pa) of length 2 m and cross-section 2 * 10^-6 m^2 is stretched by a force of 200 N. The extension is:

A. 0.5 mm **B.** 1 mm **C.** 2 mm **D.** 0.2 mm

____ **24.** The fulcrum of a uniform 20 kg seesaw 4.0 m long is located 2.5 m from one end. A 30 kg child sits on the long end. What weight must a person sitting on the other end exert to balance the seesaw? (g = 10 m/s^2)

A. 750 N **B.** 850 N **C.** 433 N **D.** 567 N

____ **25.** A body starts from rest and accelerates uniformly at 3 m/s^2 for 4 seconds. What is its final velocity?

A. 7 m/s **B.** 12 m/s **C.** 9 m/s **D.** 15 m/s

____ **26.** For a freely falling object near the Earth's surface, its acceleration is:

A. Increasing with time **B.** Constant

C. Zero **D.** Decreasing with time

____ **27.** If a body starts from rest under uniform acceleration, the displacement covered is:

A. Proportional to time **B.** Proportional to square of time

C. Proportional to cube of time **D.** Independent of time

____ **28.** Two cars A and B are moving along a straight road in the same direction at 54 km/h and 36 km/h respectively. What is the relative velocity of car A with respect to car B?

A. 18 m/s **B.** 5 m/s **C.** 90 m/s **D.** 45 m/s

____ **29.** A car starts from rest and accelerates uniformly at 5 m/s^2. How long will it take to reach a speed of 50 m/s?

A. 5 s **B.** 10 s **C.** 15 s **D.** 20 s

____ **30.** What is the diagnostic purpose of an electroscope in electrostatics?

A. To identify whether a body is charged or neutral
B. To identify whether a body is an electrical conductor or insulator
C. To identify whether the sign of charge is positive or negative
D. All of the above can be tested using an electroscope

____ **31.** How much tensile force is needed to stretch a 1 m wire of cross-sectional area 1 * 10^-6 m^2 by 1 mm if Young's modulus is 2 * 10^11 Pa?

A. 100 N **B.** 200 N **C.** 50 N **D.** 400 N

____ **32.** The slope of a velocity versus time graph represents:

A. Displacement **B.** Acceleration

C. Instantaneous velocity **D.** Time interval

____ **33.** A vehicle travelling at 20 m/s decelerates uniformly to complete rest in 5 seconds. What is its acceleration?

A. -4 m/s^2 **B.** -5 m/s^2 **C.** -3 m/s^2 **D.** -2 m/s^2

____ **34.** For a body moving with uniform acceleration, its velocity-time graph is:

A. A straight line passing through the origin
B. A straight line with constant slope equal to acceleration
C. A parabolic curve
D. A horizontal straight line

____ **35.** A car moving at 20 m/s applies brakes and comes to a stop in 10 seconds. What total distance does it cover before stopping?

A. 100 m **B.** 150 m **C.** 200 m **D.** 250 m

____ **36.** Which of the following physical quantities is completely dimensionless?

A. Tensile stress **B.** Strain

C. Young's modulus **D.** Tensile force

____ **37.** A freely falling stone released from rest falls for 3 seconds. How far does it fall? (g = 10 m/s^2)

A. 30 m **B.** 45 m **C.** 50 m **D.** 60 m

____ **38.** A body possesses a net negative charge of -6.4 nC. How many electrons are in excess? (Elementary charge e = 1.6 * 10^-19 C)

A. 4.0 * 10^10 electrons in excess **B.** 4.0 * 10^10 electrons in deficit

C. 4.0 * 10^13 electrons in excess **D.** 4.0 * 10^13 electrons in deficit

____ **39.** A cylindrical steel rod of length L = 2 m and cross-sectional area A = 1 cm^2 is subjected to a tensile force F = 10,000 N. The tensile stress in the wire is:

A. 10^8 Pascal **B.** 10^6 Pascal **C.** 10^7 Pascal **D.** 10^5 Pascal

____ **40.** For the steel rod above with Y = 2 * 10^11 Pa, what are the strain and elongation of the wire?

A. Strain = 5 * 10^-4, Elongation = 1 mm
B. Strain = 2 * 10^-4, Elongation = 2 mm
C. Strain = 1 * 10^-3, Elongation = 0.5 mm
D. Strain = 5 * 10^-3, Elongation = 10 mm